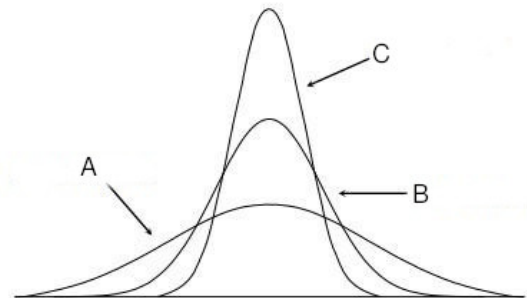


[N.B. – Answer all the questions. Each question carries ONE mark. Block fully, with a black ball- point pen, the circle of the letter that stands for the correct/best answer in the “Answer sheet” for the Multiple Choice Questions Examination.]

Candidates are asked not to leave any mark or spot on the question paper.

1. Which curve is platykurtic?



- (a) A (b) B (c) C (d) None
2. The Range of Karl Pearson's measure of skewness –
(a) (0, 1) (b) (-1, 1) (c) (-3, 3) (d) (0, ∞)
3. Which can be used to measure dispersion?
(a) μ'_2 (b) μ_1 (c) μ_2 (d) μ'_1
4. For a mesokurtik distribution, $\beta_2 = -$
(a) 0 (b) -3 (c) 3 (d) 1
5. Which of the following has the strongest correlation?
(a) -0.92 (b) 0.67 (c) 0.50 (d) 0.91
6. The lowest possible value of the correlation coefficient —
(a) 1 (b) 0 (c) $-\infty$ (d) -1
7. Which of the following is most suitable for comparing the sales of five different products over a single year?
(a) Histogram (b) Pie Chart (c) Simple Bar Diagram (d) Frequency Curve
8. Which statement is true about secondary data?
i. Already published
ii. Economical
iii. Always up-to-date
Which one is correct?
Which one is correct?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
9. Which of the following is an example of cyclic variation in a time series?
(a) Boom and recession phases in an economy
(b) Increase in electricity consumption during summer
(c) High demand for umbrellas during the rainy season
(d) Sudden decline in stock prices due to a pandemic
10. Which measure of trend is subjective?
(a) Semi-average method (b) Graphical method
(c) Moving average method (d) None of the above

11. Which of the following is an example of an infinite population?
- (a) Patients in a hospital (b) Water molecules in the ocean
(c) Cars on a highway (d) Students in a university
12. How many measurement scales are there?
- (a) 2 (b) 3 (c) 4 (d) 5
13. Given $z_1 = 3$, $z_2 = 0$, $z_3 = -3$, and $z_4 = 2$, what is the value of $\sum_{i=1}^4 (z_i^2 + 5)$?
- (a) 30 (b) 40 (c) 35 (d) 45

Answer the next THREE questions based on the following information.

The weights of 120 fruits were recorded and this frequency distribution was constructed.

Weight (grams)	0-50	50-100	100-150	150-200
No. of Fruits	30	35	25	30

14. How many fruits weigh at least 100 grams?
- (a) 55 (b) 50 (c) 60 (d) 65
15. How many fruits weigh less than 100 grams?
- (a) 68 (b) 70 (c) 65 (d) 50
16. What percent of fruits weigh between 50 and 150 grams?
- (a) 50% (b) 55% (c) 60% (d) 75%
17. Which measure is not used in determining skewness?
- (a) Arithmetic Mean (b) Geometric Mean (c) Median (d) Mode
18. The following are values of arithmetic, geometric, and harmonic mean, respectively
- i. {10, 12, 14}
ii. {10, 10, 10}
iii. {10, 8, 6}
- Which one is possible?
- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii
19. Let $Y = aX + b$, where a and b are constants. If $\bar{X} = 8$ and $\bar{Y} = 30$, and $a = 3$, $b = ?$.
- (a) 3 (b) 6 (c) 9 (d) 12
20. If $Q_1 = 25$ and $QD = 10$, what is the value of Q_3 ?
- (a) 45 (b) 70 (c) 10 (d) 25
21. What is the variance of a constant c ?
- (a) 1 (b) 0 (c) -1 (d) ∞
22. For two values, the range is found to be 8. What are the values of mean deviation and standard deviation, respectively?
- (a) (2,4) (b) (4,4) (c) (4,8) (d) (8,8)
23. Which of the following are true about variance?
- i. It is always non-negative
ii. It is measured in squared units of the original data
iii. It becomes zero when all observations are equal
- Which one is correct?
- (a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

Answer the next two questions based on the following information

The first and second moments of a distribution around 2 are 13 and 195, respectively.

24. What is the arithmetic mean?

(a) 15 (b) 12 (c) 11 (d) 26

25. What is its variance?

(a) 20 (b) 21 (c) 26 (d) 25